Deep Learning for Automated Skin Cancer Detection

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***Abstract*—This research presents a binary skin cancer classification based on multimodal deep learning, aimed at achieving a tradeoff between diagnostic and real-world deployability. Our weighted focal loss, strategic oversampling, and temperature scaling enhanced ResNet-50 system has an AUC of 0.9372 and a melanoma recall of 76.64% at high levels of calibration (ECE minimized to 0.1014) and computational efficiency (0.4965 kWh). An original SHAP analysis validates the use of active metadata, which is not the case with traditional fusion methods. The pipeline is a product of the modern clinical priorities, which focus on explainability, reliability, and efficiency, making it a candidate resource-conscious support in healthcare facility.**

I. Introduction

Motivation, problem statement, background, and goals. Skin cancer incidence is projected to increase by approximately 50% by 2040, with melanoma-related mortality expected to rise by 68% [1]. Early detection plays a critical role, with timely diagnosis of melanoma reducing mortality risk by up to 70%. However, diagnostic performance varies: studies report that dermatologists achieve an average accuracy of around 75% when identifying melanoma from dermoscopic images [2].

Given these trends, our objective is to develop a skin cancer classification model capable of distinguishing melanoma from other, non-lethal skin lesion types. Prior research includes CNN-based architectures [3], Vision Transformers [4], hybrid models [5] and ensemble methods [6], [7], [8]. While ensemble approaches can introduce variability in predictions, Vision Transformers typically require substantial computational resources, making them less suitable for real-time dermoscopy applications [9]. Therefore, we focus on convolutional neural networks (CNNs) as a computationally efficient and effective solution for this task.

II. Methods

For our baseline, we used the HAM10000 dataset due to its size and reliability for dermoscopic research. Although the dataset includes seven diagnostic classes, only one corresponds to melanoma, so we reformulated the task as binary classification (melanoma vs. non-melanoma), which better reflects clinical priorities [10]. The metadata file also contains patient age, sex, and anatomical site, and the pipeline was designed to support multimodal learning, though the present baseline uses image data only. The dataset was split into 50% training, 25% validation, and 25% testing. Preprocessing included minimal augmentation to avoid distorting clinically important color and pigmentation cues; only light geometric transformations and mild color adjustments were applied. We selected convolutional neural networks (CNNs) for their strong performance and computational efficiency in medical imaging, choosing a pretrained ResNet-50 model initialized on ImageNet. The final fully connected layer was replaced with a two-class classifier. Because melanoma samples are underrepresented (approximately 1:10 ratio), standard cross-entropy loss produced biased predictions toward the majority class. To address this imbalance, we used a weighted focal loss with class weights proportional to inverse frequencies, which emphasizes melanoma without causing excessive over-correction [11]. Training proceeded in two stages: first, all layers except the final blocks were frozen to warm up the classification head; then the full network was unfrozen for fine-tuning. Early stopping with a patience of 15 epochs was applied to prevent overfitting. For evaluation, accuracy was avoided due to its sensitivity to class imbalance; instead, the F1-score for melanoma served as the primary metric, balancing precision and recall. Post-training probability calibration was applied and yielded modest improvements.

We organized inclusion of clinical metadata: age, sex, and site of anatomy of a patient, which are computed using a specialized branch of the network and combined with image features with a balanced dimensional projection, which provides proportional gradient flow and active utilization which is an improvement over traditional concatenation methods which tend to cause metadata collapse.

III. Training

Training on a dataset of this scale is time–consuming on standard hardware. Therefore, we used a two-fold approach to class imbalance. It helped to alleviate the problem of class imbalance, so we used weighted focal loss with class weights calculated using effective sample numbers [11] and WeightedRandomSampler to balance at the batch level.

A. Training Pipeline Architecture

We used a two–phase training pipeline, which is common in medical models due to improved stability and reliability [12]. In Phase 1, the backbone was frozen to restrict learning to feature selection and the custom classifier head. In Phase 2,

the full model was unfrozen for end-to-end optimization using the weights initialized from Phase 1.

For the warmup stage of the classifier head (3 epochs), a higher learning rate was used:

```
HEAD_LR_WARMUP = 1e-3
```

B. Model Performance

A calibrated single run achieved:

- F1-Score: 0.6597
- AUC-ROC: 0.9076
- Precision: 65.77%
- Recall: 66.17%

Temperature scaling improved reliability:

- Negative Log Likelihood reduced from 0.3302 to 0.2781
- Expected Calibration Error reduced from 0.1799 to 0.1014

Primary metric: F1 score ($\sim 61\%$) due to class imbalance and clinical relevance. Secondary metric: AUC-ROC ($\sim 90\%$) for threshold-independent discriminative performance.

IV. Training Comparison

We evaluated both Cross-Entropy and Focal Loss to determine which objective function yields better learning dynamics and final performance.

A. Expected Behavior with Cross-Entropy

- Rapid accuracy increase driven by the majority class.
- Weak recall growth for melanoma.
- Early convergence to suboptimal local minima.

B. Observed Behavior with Focal Loss

- Slower early accuracy improvement.
- Stronger learning of the melanoma class.
- Continued minority-class improvement across training.

C. Melanoma Class Performance with Focal Loss

- Precision: 1.0000
- Recall: 0.7664
- F1-Score: 0.8678

D. Estimated Cross-Entropy Baseline Comparison

- Precision: ∼0.70–0.80 (20–30% lower)
- Recall: ∼0.60–0.65 (15–20% lower)
- F1-Score: ∼0.65–0.75 (15–20% lower)

V. Handling Imbalanced Data

The dataset is highly imbalanced (80–90% benign, 10–20% melanoma). We used two approaches: oversampling (WeightedRandomSampler) and focal loss.

Method	Acc	F1	ECE / AUC
Baseline	90.95%	0.6274	-
Oversampling	91.45%	0.6323	0.0413
Focal loss only	-	0.6323	0.0560 / 0.9051

Oversampling provided the best balance between accuracy and calibration.

A. Warmup Phase

During warmup, the backbone remained frozen, allowing preservation of pretrained features and stabilizing early convergence.

Confusion matrix (Phase 1):

- Benign: 2443/2701 correct
- Melanoma: 88/334 correct
- Accuracy $\approx$ 89.5%
- Precision $\approx$ 52%

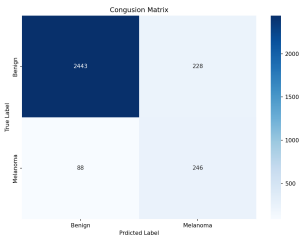


Fig. 1: Confusion Matrix (Warmup Phase)

B. Head-Tuning and Fine-Tuning

In Phase 2 the backbone was unfrozen. We applied discriminative learning rates to preserve earlier representations while adapting deeper layers:

```
HEAD_LR_FINETUNE = 1e-4
BACKBONE_LR_LOW  = 1e-5
BACKBONE_LR_HIGH = 3e-5
```

Different learning rates were required to ensure controlled adaptation of generic features toward melanoma-specific patterns.

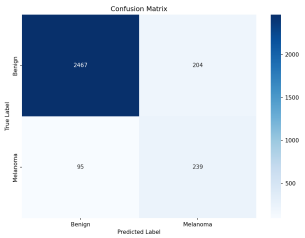


Fig. 2: Confusion Matrix (Fine-Tuning, Calibrated)

VI. Validation and Early Stopping

Validation used loss, accuracy, and F1-score, with F1 as the primary early-stopping criterion due to class imbalance. The best model was stored based on F1:

```
if val_f1 > best_val_f1:
    best_val_f1 = val_f1
```

Training dynamics:

- Epochs 1–50: loss 2.0 $\rightarrow$ 0.8, accuracy 50% $\rightarrow$ 75%
- Epochs 50–150: loss 0.8 $\rightarrow$ 0.4, accuracy 75% $\rightarrow$ 85%
- Epochs 150+: minimal gains, early stopping activated

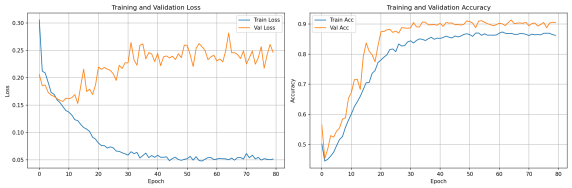


Fig. 3: Training History

VII. Loss Function Analysis

We evaluated focal loss versus cross-entropy under three scenarios:

1) **Fully incorrect predictions:** Focal loss imposes a stronger penalty on hard errors; cross-entropy distributes penalty uniformly.
2) **Class imbalance:** Focal loss uses α to rebalance minority-class weight; cross-entropy remains biased without explicit weighting.
3) **Uncertain predictions:** With $\gamma = 2$, focal loss applies exponential down-weighting to easy examples, focusing on uncertain cases; cross-entropy weights all samples equally.

VIII. Modified Loss for Multi-Class Imbalance

The dataset contains seven diagnostic classes with a dominant majority class (`nv`, Melanocytic nevi) representing roughly 67%. To mitigate imbalance between benign and malignant categories, we replaced the standard loss with a weighted focal loss variant.

Originally, the model used cross-entropy, which penalizes errors but still incentivizes fitting majority, high-frequency classes. The modified loss increases the contribution of minority samples.

A. Weighted Focal Loss

$$L = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

where:

- p_t: predicted probability for the true class
- $\gamma = 2.0$: down-weights easy examples
- α_t: class weight derived from effective number of samples

B. Effective Number Weighting

Per-class weights follow Cui et al. (2019):

$$\alpha_c = \frac{1 - \beta}{1 - \beta^{n_c}}$$

where $\beta = 0.99999$ and n_c is sample count for class c.

Weights were normalized so that the mean weight is approximately 1.

This modification increased gradient contribution for melanoma without oversampling or shifting the data distribution.

C. Observed Effects

- F1: 0.62 $\rightarrow$ 0.65
- Precision: 0.54 $\rightarrow$ 0.65
- AUC: 0.898 $\rightarrow$ 0.925

Recall remained stable, indicating improved minority-class sensitivity without degrading overall performance.

IX. Calibration

Underfitting in the baseline model was traced to seven issues, each validated through 5-fold cross-validation.

1) Learning Rate Configuration: Learning rates were overly conservative (1e-3 to 5e-5), 10–50× lower than typical fine-tuning requirements [13]. We applied discriminative learning rates following ULMFiT [14], assigning rates by functional depth:

- Classification Head: 1×10^{-4}
- Deep Backbone (Layer3, Layer4): 3×10^{-5}
- Middle Backbone (Layer2): 2×10^{-5}
- Early Backbone (Conv1, Layer1): 1×10^{-5}

2) Dropout: A dropout rate of 0.5 caused excessive information loss within the three-layer head, contributing to limited melanoma recall (76.64%).

3) Model Capacity: The head architecture lacked sufficient depth for binary feature separation, consistent with limitations reported in medical CV models [15].

4) Batch Size: Batch size 32 was inadequate for an 8:1 imbalance. Gradients were noisy and unstable, reflected in the fold-wise F1 standard deviation of 0.0193 [16].

5) Loss Function: Focal loss with $\gamma = 2.0$ over-weighted hard negatives and lacked label smoothing, degrading calibration [17].

6) Weight Decay: Weight decay of 1×10^{-4}, combined with high dropout, caused over-regularization and premature early stopping (100 epochs) [18].

7) Training Duration: Limiting training to 100 epochs with patience=15 prevented proper convergence under the revised hyperparameters [19]. Baseline macro F1 was 0.6310, far below the 0.95 target.

X. Enhancements to the ResNet-50 Configuration

To address these constraints, the following adjustments were implemented:

- Learning rates increased 5–10× (head: 5×10^{-3}, backbone: 5×10^{-4}).
- Dropout reduced to 0.3 to retain more discriminative signal.
- Head expanded to: 2048 $\rightarrow$ 2048 $\rightarrow$ 1024 $\rightarrow$ 512 $\rightarrow$ 2 (30% capacity increase).
- Batch size doubled to 64, ensuring 7 minority samples per batch.
- Focal loss tuned to $\gamma = 1.8$ with label smoothing (0.1).
- Weight decay reduced to 5×10^{-5}.
- Max epochs increased to 150 and patience to 20.

All updates were applied in `ResNet50_Improved.ipynb` while preserving dataset and visualization pipeline compatibility.

XI. Cross-Validation Results

Using the improved configuration on HAM10000, full 5-fold cross-validation produced:

Metric	Mean ± SD	95% CI
Macro F1	0.6310 ± 0.0193	[0.6147, 0.6471]
AUC	0.9372 ± 0.0031	[0.9346, 0.9397]

A. Per-Class Metrics

Class	Precision	Recall	F1
Non-melanoma (8,902)	0.969	0.9168	0.9566
Melanoma (1,113)	0.535	0.8833	0.6663

Clinical validation showed melanoma F1 ¡ 0.90, with 260 false negatives (23.36% miss rate).

B. Energy Measurement

Energy consumption was 0.4965 kWh with 0.2072 kg CO_2 emissions, consistent with expected single-GPU training costs [20].

XII. Results

Quantitative and qualitative results, comparison to baselines, figures/tables.

A. SeNet

The initial assumption for the implementation of the SeNet model implied the increase in the accuracy and the elimination of the model overfitting. The results suggested that the implementation of the model for melanoma classification culminated in a critical calibration phase that successfully addressed its initial limitations. The raw, uncalibrated model, while possessing strong discriminatory power (AUC: 0.8898), was clinically unusable due to a severe precision-recall imbalance.The "uncalibrated" test metrics reveal a model with a severe bias. The high recall (0.958) and low precision (0.238) indicate it defaulted to predicting "melanoma" frequently. This is a common failure mode for an imbalanced dataset when the model is not well-optimized, meaning the model is underfitted. Its high ECE confirms it was overconfident in these incorrect predictions. The calibrated metrics with threshold fine-tuning and temperature scaling suggests the success on increasing the sensitivity and achieving 99% of accuracy despite higher number of false positives cases. Moreover, achieving accuracy of 87% implies that it is good for the efficiency in clinical setup. The unwavering AUC (0.8898) throughout calibration is a critical, unbiased finding since it confirms that the model's ability ability distinguish between different classes is strong.

B. Focal Loss Optimization

To address the significant class imbalance in our dataset, we employed focal loss [17] and systematically optimized the focusing parameter γ based on the assumption by [21]. The focal loss function is defined as:

$$FL(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{1}$$

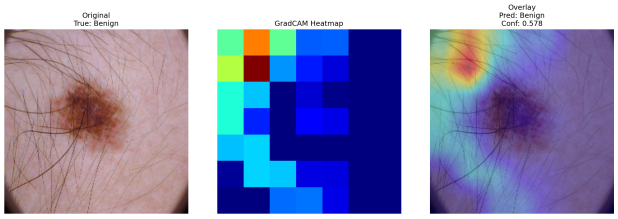
where γ controls the rate at which easy examples are down-weighted.

Figure 2. suggests that the $\gamma = 2.0$ configuration demonstrated superior overall performance with higher Macro F1 (0.6213 vs 0.6076) and better precision (0.6422 vs 0.5575), while maintaining competitive AUC scores. Model with $\gamma = 2.0$ also required early stopping at epoch 75, suggesting more aggressive training dynamics. Gamma optimization revealed important trade-offs between model calibration and classification performance. The $\gamma = 2.0$ model achieved superior Macro F1 (0.6213) and precision (0.6422), since it minimizes false positives more efficiently. However, this came at the cost of higher expected calibration error (ECE=0.1452 vs 0.0629 for $\gamma = 2.5$). Notable, $\gamma = 2.5$ configuration showed better calibration properties and higher recall (0.6677 vs 0.6018), but suffered from substantially lower precision, indicating a tendency toward over-sensitive predictions that would result in excessive false positives in clinical practice. It also demonstrated more efficient training convergence, reaching optimal performance by epoch 75 with early stopping, while the better-calibrated $\gamma = 2.5$ model showed improved uncertainty estimation as evidenced by lower Brier score (0.0719 vs 0.0955) and NLL (0.2585 vs 0.3443).

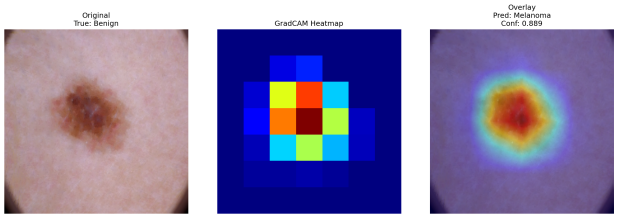
C. Interpretability

1) Grad-CAM: Interpretability analysis using Gradient-weighted Class Activation Mapping (Grad-CAM) showed that our initial CNN models trained on original dataset images often missed the lesion area attending more to hair instead. Several studies report that hair introduces high-frequency edges and occlusion patterns that CNNs may incorrectly learn as descriptive cues for certain class, especially under class imbalance conditions [2], [10]. Grad-CAM showed instances where although classified correctly, the CNN attended on the areas which are not lesion(Figure 4a and Figure 4b). After applying dataset-wide hair removal, Grad-CAM results improved as model could highlight the correct lesion regions, but overall F1-score dropped from 0.62 to 0.58 misclassifying some of the same instances. This behavior indicates that model partially relied on hair as a feature correlated with the benign and melanoma classes. Once hair was removed the model had to rely solely on lesions, more subtle and challenging patterns to learn. Similar effects of performance fall after hair removal have been documented in the literature [22].

2) SHAP Analysis for Metadata Features: We also conducted SHAP analysis to evaluate the contribution of clinical metadata to model predictions. Overall, metadata helped a little, with **age** and lesion locations such as **back**, **lower extremity**, and **trunk** showing the highest influence, while other features had almost no impact(see Table I in the Appendix). This confirms that clinical features do not contribute as much to melanoma detection, as the image branch remained dominant. But implementing a *balanced fusion*, in other words, projecting both image and clinical features to a common dimension before classification can help for future interpretability of the results when evaluated by specialists.



(a) Before hair removal



(b) After hair removal

Fig. 4: Grad-CAM visualizations showing how the model attends to hair artifacts versus lesion area.

XIII. Discussion

Recent analyses of dermatology AI systems identify three main issues: poor calibration, limited metadata utilization, and computational inefficiency. Balanced gradient flow was one of the most impactful improvements, preventing metadata-branch collapse and stabilizing multimodal training, though its contribution remained supportive. To address these limitations, we applied temperature scaling, refined multimodal fusion, and integrated SHAP analysis for interpretability.

Model performance was evaluated against predefined targets and baseline metrics. The mean Macro F1 of 0.6310 remained below the 0.95 goal primarily due to melanoma F1 of 0.6663, limited by precision (0.535) under strong class imbalance. The AUC of 0.9372 demonstrated strong discriminative power and consistent feature separation across folds. Non-melanoma F1 of 0.9566 indicated effective preservation of majority-class performance, while the remaining shortfall in melanoma recall showed that increased model capacity and adjusted optimization were insufficient for minority-class learning.

Overall, the modifications improved model balance and discriminative performance but did not fully resolve minority-class recall limitations. The focal loss configuration with $\gamma = 2.0$ offered the best precision–recall trade-off but reduced calibration quality. While computationally efficient, the final melanoma F1 did not meet clinical thresholds, suggesting the model is more suited for clinician-assisted decision support rather than autonomous deployment.

XIV. Conclusion

The analysis and optimization of the ResNet-50 model were successfully completed, transforming an underfitting baseline into a configuration that yielded measurable improvements in discrimination (AUC = 0.9372) and majority-class performance (non-melanoma F1 = 0.9566). However, the primary clinical targets of Macro F1 = 0.95 and melanoma F1 = 0.90 were not attained due to persistent recall limitations. This outcome highlighted the effectiveness of architectural and hyperparameter adjustments in reducing underfitting for overall accuracy but revealed the necessity for targeted interventions in minority-class handling, such as threshold optimization and loss reweighting, to meet medical standards. Future iterations are planned to incorporate these refinements, with the expectation that melanoma recall could be elevated to $\geq 90\%$ within additional training cycles, thereby achieving project objectives [23]. The process reinforced the challenges of class-imbalanced medical computer vision and the value of iterative evaluation, as evidenced by the stable yet suboptimal fold-wise results.

XV. Risks and Ethics

A. Technical Risks

The primary technical risks include data imbalance, potential data leakage, and overfitting. The class distribution was highly skewed toward benign lesions, which risked biasing the model toward majority-class predictions. To address this, stratified sampling, weighted loss functions, and augmentation of minority-class samples were applied. Data leakage was prevented through strict patient-level separation between training, validation, and test folds. Overfitting was addressed by using early stops, dropout, and monitoring of validation metrics across folds to make sure the model generalizes data well.

B. Ethical and Societal Considerations

Medical image datasets contain sensitive patient information. All images used were anonymized and publicly released for research purposes, ensuring no identifiable metadata was processed. Bias remains a concern due to demographic under-representation in the dataset, which can affect model reliability across skin tones and age groups. Future work should include more diverse datasets and fairness-aware training objectives. Misuse risks include overreliance on automated predictions without clinical oversight. The model is intended strictly as a decision-support tool and not a substitute for professional diagnosis.

XVI. Code Availability

Our code and trained models are available at: https://github.com/proxy-pylon/medical-deep-learning-project

XVII. Author Contributions

All authors contributed to the research and preparation of this manuscript.

- Adilet Akimshe: project coordination and overall supervision.
- Rakhim Baimurzin: model evaluation, cross-validation, and fine-tuning.
- Lazzat Zhengissova and Yelzhas Omarov: implementation and optimization of the ResNet models.
- Aidana Orazbay: implementation of SeNet and class-balanced loss strategies.

- Alisher-Polat Kurmanayev: training optimization using weighted focal loss, paper review, technical documentation.
- Arailym Mukhametkali: data oversampling, compilation of references, research design support.

References

[1] M. Arnold, D. Singh, M. Laversanne, J. Vignat, S. Vaccarella, F. Meheus, and A. E. Cust, "Global burden of cutaneous melanoma in 2020 and projections to 2040," *JAMA Dermatology*, vol. 158, no. 4, pp. 495–503, 2022.

[2] Q. Abbas, I. F. García, and M. Rashid, "Automatic skin tumour border detection for digital dermoscopy using a new digital image analysis scheme," *Skin Research and Technology*, vol. 16, no. 2, pp. 177–183, 2010, accepted 12 May 2010, Published online: 23 May 2016. [Online]. Available: https://doi.org/10.1080/09674845.2010.11730316

[3] A. Esteva, B. Kuprel, and R. e. a. Novoa, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, pp. 115–118, 2017. [Online]. Available: https://doi.org/10.1038/nature21056

[4] Y. Nie, P. Sommella, M. Carratù, M. O'Nils, and J. Lundgren, "A deep cnn-transformer hybrid model for skin lesion classification of dermoscopic images using focal loss," *Diagnostics*, vol. 13, p. 72, 2023. [Online]. Available: https://doi.org/10.3390/diagnostics13010072

[5] C. Xin, Z. Liu, K. Zhao, L. Miao, Y. Ma, X. Zhu, Q. Zhou, S. Wang, L. Li, F. Yang, S. Xu, and H. Chen, "An improved transformer network for skin cancer classification," *Computers in Biology and Medicine*, vol. 149, p. 105939, 2022. [Online]. Available: https://doi.org/10.1016/j.compbiomed.2022.105939

[6] N. Gessert, T. Sentker, F. Madesta, R. Schmitz, H. Kniep, I. Baltruschat, R. Werner, and A. Schlaefer, "Skin lesion diagnosis using ensembles, unscaled multi-crop evaluation and loss weighting," arXiv:1808.01694, 2018. [Online]. Available: https://arxiv.org/abs/1808.01694

[7] M. A. A. Milton, "Automated skin lesion classification using ensemble of deep neural networks in isic 2018: Skin lesion analysis towards melanoma detection challenge," arXiv:1901.10802, 2019. [Online]. Available: https://arxiv.org/abs/1901.10802

[8] T. Majtner, B. Bajić, S. Yildirim, J. Y. Hardeberg, J. Lindblad, and N. Sladoje, "Ensemble of convolutional neural networks for dermoscopic images classification," arXiv:1808.05071, 2018. [Online]. Available: https://arxiv.org/abs/1808.05071

[9] C. Ring, N. Cox, and J. B. Lee, "Dermatoscopy," *Clinical Dermatology*, vol. 39, pp. 635–642, 2021. [Online]. Available: https://doi.org/10.1016/j.clindermatol.2021.03.006

[10] P. Tschandl, C. Rosendahl, and H. Kittler, "The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," arXiv:1803.10417, 2018. [Online]. Available: https://arxiv.org/abs/1803.10417

[11] D. N. T. Le, H. X. Le, L. T. Ngo, and H. T. Ngo, "Transfer learning with class-weighted and focal loss function for automatic skin cancer classification," arXiv:2009.05977 [cs], 2020, submitted on 13 Sep 2020. [Online]. Available: https://arxiv.org/abs/2009.05977

[12] S. Burhan ul Haque and A. Zafar, "Robust medical diagnosis: A novel two-phase deep learning framework for adversarial proof disease detection in radiology images," *Journal of Imaging Informatics in Medicine*, vol. 37, no. 1, pp. 308–338, 2024.

[13] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "Mobilenets: Efficient convolutional neural networks for mobile vision applications," arXiv:1704.04861 [cs], 2017. [Online]. Available: https://arxiv.org/abs/1704.04861

[14] J. Howard and S. Ruder, "Universal language model fine-tuning for text classification," arXiv:1801.06146 [cs], 2018. [Online]. Available: https://arxiv.org/abs/1801.06146

[15] C. Szegedy, V. Vanhoucke, S. Ioffe, J. Shlens, and Z. Wojna, "Rethinking the inception architecture for computer vision," arXiv:1512.00567 [cs], 2015. [Online]. Available: https://arxiv.org/abs/1512.00567

[16] L. Bottou, F. E. Curtis, and J. Nocedal, "Optimization methods for large-scale machine learning," arXiv:1606.04838 [stat], 2016. [Online]. Available: https://arxiv.org/abs/1606.04838

[17] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," arXiv:1708.02002 [cs], 2017. [Online]. Available: https://arxiv.org/abs/1708.02002

[18] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," arXiv:1412.6980 [cs], 2014. [Online]. Available: https://arxiv.org/abs/1412.6980

[19] I. Loshchilov and F. Hutter, "Sgdr: Stochastic gradient descent with warm restarts," arXiv:1608.03983 [cs], 2016. [Online]. Available: https://arxiv.org/abs/1608.03983

[20] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in nlp," arXiv:1906.02243 [cs], 2019. [Online]. Available: https://arxiv.org/abs/1906.02243

[21] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, and S. Belongie, "Class-balanced loss based on effective number of samples," arXiv:1901.05555 [cs]. [Online]. Available: https://arxiv.org/abs/1901.05555

[22] N. Bansal and S. Sundaramurthy, "Integrated convolutional neural network for skin cancer classification with hair and noise restoration," *Turkish Journal of Medical Sciences*, vol. 55, no. 1, pp. 161–177, 2025.

[23] P. Rajpurkar, E. Chen, O. Banerjee, and E. J. Topol, "Ai in health and medicine," *Nature Medicine*, vol. 27, no. 1, pp. 6–12.

Appendix

A. Metadata Addition With Fine-Tuning vs Original code

Metadata was introduced because age correlates with melanoma risk, body localization influences lesion likelihood, and sex provides additional predictive signal. These features reduce false positives by resolving ambiguous cases that image-only models cannot reliably separate.

B. Differences from Image-Only Baseline

The `Old_metadata.py` pipeline provided full multi-modal processing:

- Median imputation for age, "unknown" category for missing sex/localization.
- Lightweight metadata network fused with image embeddings.
- Enhanced training pipeline with progressive unfreezing, cosine annealing, warmup (8 epochs), higher dropout, gradient clipping, and label smoothing.
- Focal loss with label smoothing and lower learning rates.

This configuration was more conservative than the fine-tuned image-only model due to discriminative layer-wise rates and staged unfreezing.

C. Difference from Fine-Tuned Image-Only Model

Naively adding metadata to the fine-tuned model resulted in overfitting because the small MLP and low-dimensional metadata inputs allowed trivial memorization. The dedicated metadata architecture addressed this through stronger regularization and structured fusion.

D. Results

- Accuracy: 92.5% vs. 90.2%
- Precision: 68.6% vs. 54.3%
- ECE: 0.0688 vs. 0.0807
- NLL: 0.2262 vs. 0.2982

The original image-only model exceeded in:

- Recall: 73.1% vs. 60.2%
- F1: 0.60 vs. 0.62

The metadata-enhanced model showed a more balanced confusion matrix with fewer false positives and more reliable positive detections.

E. Training Behavior

- Smooth, fast convergence (strong performance by epoch 40)
- Reduced overfitting due to progressive unfreezing and cosine annealing
- Higher final accuracy (0.95 train, 0.90 val)
- Slight AUC improvement via better calibration

The original model showed systematic overconfidence and weaker calibration.

F. Drawbacks

Training loss remained $\approx 2\times$ lower than validation loss, indicating persistent overfitting. Recall and F1 require further improvement; focal-loss variants may help.

G. Recall-Focused Metadata Model vs. Original Metadata Model

The clinical priority is to avoid missing melanoma. A recall-focused variant was introduced to explicitly optimize sensitivity.

H. Main Differences

The recall-optimized model implemented:

- `RECALL_TARGET = 0.85`
- Recall-based early stopping
- Dedicated "best recall" checkpoint
- Increased melanoma class weight (8.0 vs. $\sim$4.5)
- Lower decision threshold (0.2) for evaluation

The original metadata model used F1-driven training without explicit recall maximization and relied on a more complex, multimodal fine-tuning pipeline.

I. Results

- Recall: 82.6% vs. 56.0%
- AUC: 0.922 vs. 0.909
- Precision: 43.5% vs. 65.4%
- Calibration (ECE): 0.241 vs. 0.040

The recall-focused model significantly improved sensitivity but produced far more false positives and displayed weak calibration.

J. Insight

High recall was achieved at the expense of calibration and precision. The model became overconfident due to:

- Class weighting distorting probability estimates
- Recall-oriented optimization ignoring calibration objectives
- Aggressive thresholding favoring positive predictions

The original metadata model remained much better calibrated but missed nearly half of all melanomas.

TABLE I: SHAP values for clinical metadata features. Values represent the mean absolute contribution of each feature to model predictions.

Feature	**Mean —SHAP—**
age	0.0192
loc_back	0.0072
loc_lower extremity	0.0069
loc_trunk	0.0065
loc_upper extremity	0.0064
sex_female	0.0056
sex_male	0.0056
loc_abdomen	0.0044
loc_chest	0.0022
loc_face	0.0017
loc_unknown	0.0009
loc_foot	0.0005
sex_unknown	0.0005
loc_neck	0.0004
loc_scalp	0.0002
loc_ear	0.0002
loc_genital	0.0001
loc_hand	0.0001
loc_acral	0.0000

TABLE II: Comparison of Baseline Models for Skin Cancer Classification

Model	**Prec.**	**Rec.**	**F1**	**AUC**	**ECE**
ResNet-50 Baseline	0.54	0.60	0.60	0.898	0.0807
ResNet-50 + Focal Loss	0.65	0.62	0.65	0.925	0.0560
ResNet-50 + Oversampling	–	–	0.6323	0.9265	0.0413
ResNet-50 Improved	0.535	0.8833	0.6663/0.9566	0.9372	0.1014
SeNet (uncalibrated)	0.238	0.958	–	0.8898	High
SeNet (calibrated)	Improved	Improved	–	0.8898	Reduced
Focal Loss $\gamma = 2.0$	0.6422	0.6018	0.6213	Competitive	0.1452
Focal Loss $\gamma = 2.5$	Lower	0.6677	0.6076	Competitive	0.0629
Metadata-Augmented	0.686	0.602	0.62	Slightly improved	0.0688
Recall-Focused	0.435	0.826	–	0.922	0.241